

GM and Google team up

This team up is evidently for developing an application for Android that will give important information about Chevrolet's upcoming car "Volt" on the driver's cell phone

Android's leaking juice!

Everybody knows Android phones are hard on batteries. Some Google employees have taken to blaming application developers for this particular flaw

For some reason the remote didn't work and we had to use the buttons on the side of the unit itself, these buttons are small and hard, and thus, hurt ones fingertips.

Another bug - 1080p, for which there is native support would appear with two black vertical strips on each side, and there was no way to remedy this. Text wasn't sharp, despite fiddling with the settings. Reducing the contrast doesn't do much - in fact the difference between contrast being set at 100 and 70 is barely noticeable.

We could see slight backlight bleed in the centre of the display. In the shade ramp test, the three most intense hues of red and green blend. This is pretty poor. In reverse text tests yellow and white on green backgrounds were quite poor, as was yellow text on a white background.

There is no separation between black and grey - this makes certain darker scenes with subtle variations in grey blend. Bright halos aren't done very well - there seems to be an orange hue to them and highlights aren't as spectacular as they're meant to be. Additionally, there is banding visible in blue-sky scenes.

There is slightly perceptible input lag in games, but it doesn't affect gameplay, unless you're playing lightning fast,

HOW WE TESTED

We divided the displays primarily on their sizes and then on the price. For this test we connected these displays to a computer and all testing was done via HDMI connectivity, which is, the most common connect used. The PC used consisted of a Core 2 Duo E6700 CPU, ASUS EAH5850 graphic card, 4 GB Corsair DDR3 RAM, ASUS P5Q3 Deluxe motherboard and 500 GB Western Digital HDD. The graphics card has a dedicated HDMI port, so no converter was used. Windows 7 Ultimate 32-bit was our OS of choice. For media playback we used Windows Media Player with the relevant codecs.

First of all, we used the Spyder3Elite to calibrate each display, and in the bargain got values for black and white levels and brightness. Maximum brightness is arrived at by setting brightness, contrast and backlighting to their maximum values, and is a look at how bright the display can get. What was interesting that our brightness differed from manufacturers rating. For nominal brightness we set brightness, contrast and backlight intensity to comfortable, normalised levels. We refrained from using a set value of 50, 50, 50 for each parameter, as different displays are relatively brighter at even the same settings. The luminance level of white and black was tested at nominal brightness levels. Before calibration we set brightness, contrast and backlighting to levels that were pleasing and uniform to the eye. The following settings were done on the TVs: Colour tones were set to neutral, or medium where applicable. Sharpening was turned off, or kept down. Colour tone

was neutral. Zero bias was used for RGB values. Noise reduction was turned off, as was any dynamic contrast and backlight settings. If a display was capable of more than 60 Hz, this was also turned off. Some displays have a cinema mode (24 frames per second), also called the 2-3 pulldown, this was also turned off. Since calibrating using the Spyder got us such diverse results, we decided to finely tune each TV to what appeared to be a better image. There is no fixed formula for calibration - some displays look too vibrant, some too washed out, some have a very low default black level, while others do not have sufficient brightness.

After calibration and logging of the values that were thrown up, we ran DisplayMate Video Edition to check the contrast, blackness level and grey-scale intensity ourselves. The grey-scale intensity is an important test, for while a display might be able to produce good blacks and intense whites, many flounder to accurately produce a wide gamut of grey - some intensities of grey tend to blend, and as the intensity comes closer to black, we see a lot of problems to display shades very close to black, but not quite black. This is of great essence with any sort of multimedia including both movies and games. If a display is measured to have poor greyscale intensity, you will not see detail in shadows and dimly lit scenes - either such areas will look washed out, or everything will appear black. DisplayMate was also used to fine-tune the calibration given by the Spyder3Elite.

The videos (1080p, 720p and DVD) we used consisted of full length

videos and action clips from the following movies: Iron Man 2, Resident Evil: Afterlife, The Human Centipede, Kingdom Of Heaven, Australia, Laws Of Attraction. The HD videos were all uncompressed Blu-Ray-format files. The DVD videos were .VOB files. For the DIVX playback via USB, we used two DIVX files, one transcoded from a .VOB file, and the other from a 720p video.

We used 1080p and 720p videos in addition to DVD videos to see how these displays handled not only native content but also non-native content, where some sort of downscaling is required. These clips were a mix of dark scenes, lots of complex colours, fast action scenes and bright daylight. For our games tests we used *S.T.A.L.K.E.R Call Of Pripyat* and *Crysis Warhead*. Games were run at their native resolution. We checked for colour, contrast, brightness and HDR of the displays, as well as pixel and backlight response. Additionally we looked at the occurrence of input lag, characterised by a lazy mouse with some delay in response - all displays have it, we rated how severe it was.

For features, we noted the presence of connectivity options, as well as the menu options available. We judged the quality and usability of buttons on the displays and their remote units, and the intuitiveness of the menu system. Scores were awarded on a scale of 1 to 10 for subjective opinions, whereas measurable values were taken for objective tests. We didn't assign weights to luminance and contrast ratio values given by the Spyder3Elite simply because a slight change in any setting impacts the scores.



twitch-based first person shooters. Good colour and sharpness in *Crysis Warhead*, but this refused to run in full screen mode. *S.T.A.L.K.E.R* ran fine in full screen mode, no issues with input lag and colours and contrast were pretty good, except that darker scenes look a little less detailed.

This display comes with

DTH built in that only needs an antenna to work. At Rs. 35,990 the 3205HD isn't a bad deal - it's slightly better than Onida's Diamond Kyrock with a couple of extra features thrown in for the same price tag. But it cannot compete with the superior quality of the MoserBaer MBIK32HD and Haier LB32R3 - which are cheaper. **d**